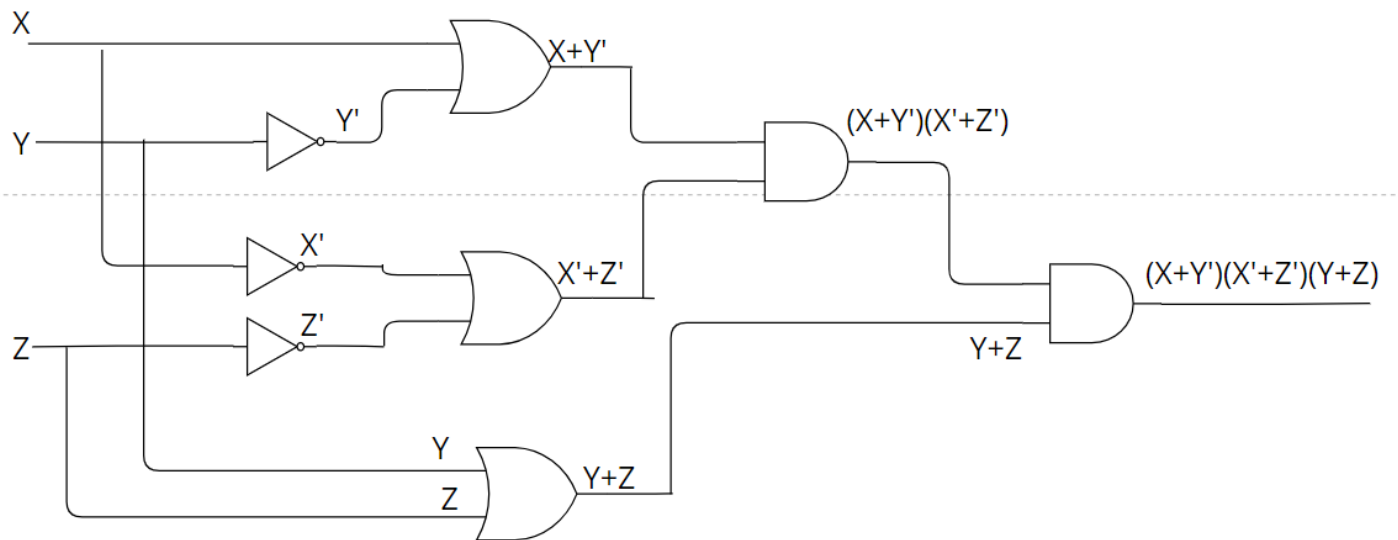
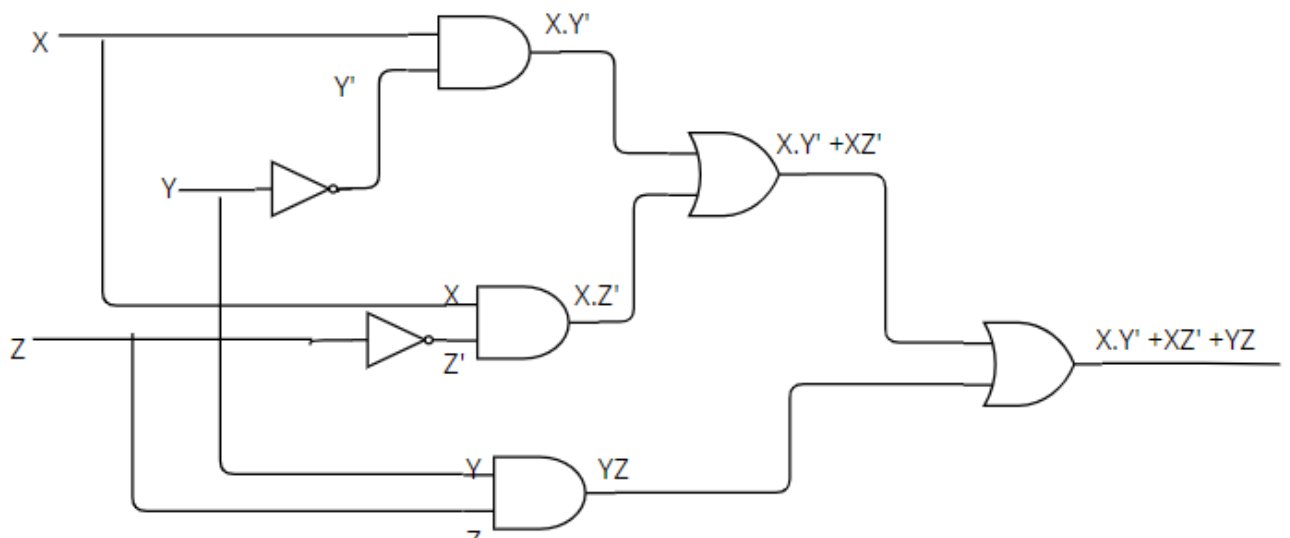


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Task 1



Task2

a. Core:

A core is an individual processing unit within a computer's CPU (Central Processing Unit). Each core can execute instructions independently, allowing for parallel processing. In a multi-core processor, multiple cores work together, enabling the computer to handle multiple tasks simultaneously, which can significantly improve performance for tasks like multitasking or running complex applications.

b. Processor:

A processor, or CPU (Central Processing Unit), is the brain of the computer responsible for executing instructions and performing calculations. In the context of a multi-core computer, the processor refers to the entire chip that contains multiple cores. The term "processor" can sometimes be used interchangeably with "CPU," but in a multi-core setup, it specifically refers to the chip that houses all the cores, which work together to process data and execute tasks efficiently.

Task 3

a. Prepare a Greeting Card:

To design and create a greeting card, a Personal Computer (PC) or a Laptop would be ideal. These types of computers are perfect for tasks that involve creativity, like using graphic design software or word processing programs to make personalized cards.

b. Smartphone:

For a smartphone, you'd actually be using a Mobile Device. Smartphones are like mini-computers that fit in your pocket, allowing you to make calls, browse the internet, run apps, and stay connected on the go.

c. Batch Data Processing in a Bank:

For handling large-scale tasks like batch data processing in a bank, a Mainframe Computer would be used. Mainframes are powerful machines designed to process massive amounts of data quickly and efficiently, making them ideal for critical financial operations.

Task 4

1. x86:

Widely used in personal computers, servers, and some mobile devices, x86 is an ISA developed by Intel. It's known for its backward compatibility and extensive use in consumer and enterprise computing.

2. ARM (Advanced RISC Machine):

ARM is a RISC (Reduced Instruction Set Computer) architecture widely used in mobile devices, embedded systems, and increasingly in laptops and servers. It's known for its power efficiency, making it ideal for battery-powered devices.

3. RISC-V:

RISC-V is an open-source Instruction Set Architecture (ISA) based on RISC (Reduced Instruction Set Computing) principles. Unlike proprietary ISAs like x86 and ARM, RISC-V is free and available for anyone to use, which has boosted its popularity in both academia and industry. Its modular and flexible design allows for customization across a range of applications, from embedded systems to high-performance computing. This openness and scalability make RISC-V an appealing and rapidly growing competitor to established architectures.

Task 5

1. X represents the Central Processing Unit (CPU).

The CPU is responsible for executing instructions and processing data.

2. Y represents the Memory (RAM). This is where data and instructions are stored temporarily while being processed by the CPU.
3. Z represents the Input/Output (I/O) Unit. This handles communication between the computer and the outside world, such as with keyboards, monitors, and other peripherals.